14/10/2026

Transfer Learning

Uses Prefrained Model's Knowledge to apply it to a different Problem but related Problem.

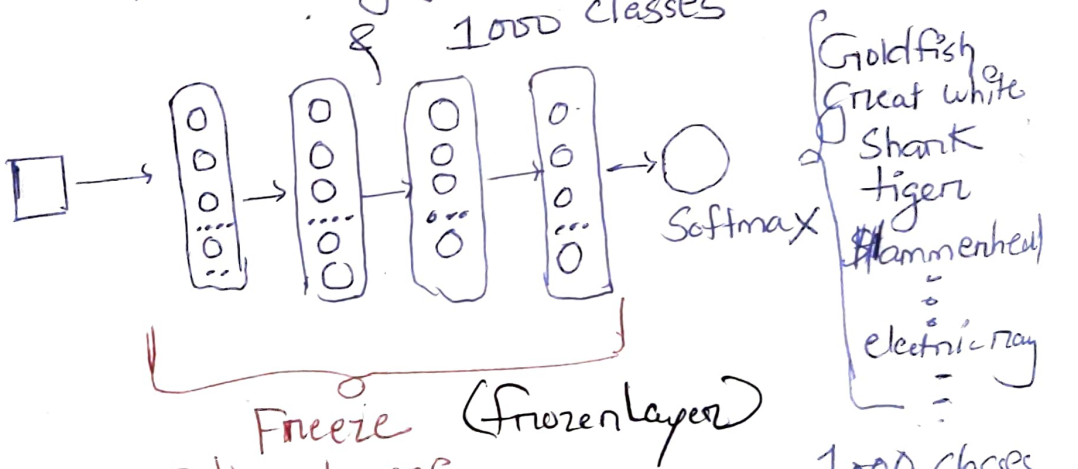
→ trained on Large datasets.

eg: Knowledge gained while learning to recognize cars could apply when trying to recognize trucks.

Transfer Learning (TF) is a research problem in machine Learning (ML) that focuses on storing Knowledge gained while solving one problem and applying it to a different but related Problem.

Eg: Googles Mobilenet V2 Model

Trained on google, 1.4 Million Images & 1000 classes



Model's weights will not change during Training

Extract feature Vector.

Tensorflow Hub

→ available many Pretrained Model

```
import numpy as np
import cv2
import PIL.Image as Image
import os
import matplotlib.pyplot as plt
import tensorflow as tf
import tensorflow_hub as hub
from tensorflow import keras
from tensorflow.keras import layers
from tensorflow.keras.models import Sequential
```

pip install tensorflow-hub → else ModuleNotFoundError

```
classifier = tf.keras.Sequential([
    hub.KerasLayers("
])
```

", input_shape = (IMAGE_SHAPE + (3,))

(224, 224) + (3,) #Numpy

→ (224, 224, 3)

```
gold_fish = Image.open("gold_fish_img.jpg").resize(IMAGE_SHAPE)
gold_fish
gold_fish = np.array(gold_fish)/255   # Normalize or Scaling Pixels.
gold_fish.shape → (224, 224, 3)
gold_fish
gold_fish[np.newaxis, ...].shape
→ (1, 224, 224, 3)
```

Because model predicts when give input as (multiple Images)

```
classifier.predict(gold_fish[np.newaxis, ...])
```

(1, 224, 224, 3) # of Image

gold_fish[np.newaxis, ...] ≡ gold_fish[np.newaxis, :, :, :]

...: Select all remaining dimensions

gold_fish[np.newaxis, ...] ≡ gold_fish[np.newaxis, :, :, :]

np.newaxis: added new dimension of length 1

If final layer does not have activation = 'softmax'

from_logit = True ↑ use

if final layer have activation = 'softmax' ⇒ from_logit = False. (default)